

SCORE:

Math 415
Modern Algebra I

NAME: _____

PRACTICE MIDTERM II

Please show your work and write only in **pen**. Notes are forbidden. Calculators, and all other electronic devices, are forbidden. Brains are encouraged, but at most one may be used per exam.

1: Please classify all group homomorphisms from $\mathbb{Z}/4\mathbb{Z}$ into $\mathbb{Z}/10\mathbb{Z}$.

Done in class.

Instructor: J. Maurice Rojas

2: Please find all group homomorphisms from $(\mathbb{Z}/17\mathbb{Z})^*$ into $(\mathbb{Z}/7\mathbb{Z})^*$.

Done in class.

3: Please prove that, for any distinct odd primes p and q , $(\mathbb{Z}/pq\mathbb{Z})^*$ is isomorphic to $(\mathbb{Z}/p\mathbb{Z})^* \times (\mathbb{Z}/q\mathbb{Z})^*$. **Hint:** Find a natural function mapping from the first group to the second that is easily provable to be an isomorphism.

Done in class.

- 4:** Suppose $A := [a_{i,j}]$ is any $n \times n$ matrix with integer entries, and $x := (x_1, \dots, x_n)$ is a vector of variables. Let us then define x^A to be the n -tuple $(x_1^{a_{1,1}} \cdots x_n^{a_{n,1}}, \dots, x_1^{a_{1,n}} \cdots x_n^{a_{n,n}})$.
- (a) Please compute $(2, 3) \begin{bmatrix} 4 & 7 \\ -2 & 5 \end{bmatrix}$.
- (b) Please prove that for any $n \times n$ matrices A and B , $x^{AB} = (x^A)^B$.
- (c) Assume $\det A = \pm 1$. Show that the function $f : (\mathbb{R}^*)^n \longrightarrow (\mathbb{R}^*)^n$ defined by $f(x) := x^A$ is a group isomorphism. **Hint:** Find a simple formula for the inverse of f using Part (b).

5: Suppose G is any group and H is a normal subgroup with G/H of cardinality 3. Please prove that the third power of any element of G lies in H .

Done in class.

6: Suppose G is any group and $H, K \leq G$. Show by an explicit example that $H \cup K$ need **not** be a subgroup of G .

Done in class.

7: Consider the elements $a := \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ and $b := \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$ of $\text{GL}_2(\mathbb{Q})$.

- (a) Please show that a and b have respective orders 4 and 3.
- (b) What is the order of ab ?

Done in class.

8:

We define the *commutator subgroup* of a group G — called $[G, G]$ — to be the intersection of all normal subgroups containing $\{aba^{-1}b^{-1} \mid a, b \in G\}$.

(a) Please prove that $G/[G, G]$ is always Abelian.

(b) Please prove that the commutator subgroup of $\mathrm{GL}_n(\mathbb{R})$ is $\mathrm{SL}_n(\mathbb{R})$.

(a): Done in class.

(b): To prove the groups are the same we proceed by proving the containments $[\mathrm{GL}_n(\mathbb{R}), \mathrm{GL}_n(\mathbb{R})] \subseteq \mathrm{SL}_n(\mathbb{R})$ and $\mathrm{SL}_n(\mathbb{R}) \subseteq [\mathrm{GL}_n(\mathbb{R}), \mathrm{GL}_n(\mathbb{R})]$.

The first containment is routine: By the multiplicativity of the determinant, it is clear that $\det(XYX^{-1}Y^{-1}) = 1$. $\mathrm{SL}_n(\mathbb{R})$ thus contains all commutators. We are not done yet, however: $[\mathrm{GL}_n(\mathbb{R}), \mathrm{GL}_n(\mathbb{R})]$ is the *intersection of all normal subgroups containing* all commutators. In particular, if we can show that $\mathrm{SL}_n(\mathbb{R})$ is a normal subgroup of $\mathrm{GL}_n(\mathbb{R})$, then we obtain $[\mathrm{GL}_n(\mathbb{R}), \mathrm{GL}_n(\mathbb{R})] \subseteq \mathrm{SL}_n(\mathbb{R})$. However, since $\mathrm{SL}_n(\mathbb{R})$ is the kernel of the homomorphism $\det : \mathrm{GL}_n(\mathbb{R}) \rightarrow \mathbb{R}$, we have the normality of $\mathrm{SL}_n(\mathbb{R})$ and we have our first containment.

The second containment is a bit more tricky: we need to show that every element of $\mathrm{SL}_n(\mathbb{R})$ is a commutator. The case $n = 1$ is trivial (since $\mathrm{SL}_1(\mathbb{R}) = \{1\}$), so let us assume henceforth that $n \geq 2$. Toward this end, for any distinct $i, j \in \{1, \dots, n\}$, let $X_{i,j}(\alpha)$ denote the $n \times n$ matrix that agrees with the $n \times n$ identity matrix in all entries, except for the (i, j) -entry, which we set to α . It is then easily checked that $X_{i,j}(\alpha)^{-1} = X_{i,j}(-\alpha)$ and $X_{i,k}(\alpha)X_{k,j}(1)X_{i,k}(\alpha)^{-1}X_{k,j}(1)^{-1} = X_{i,j}(\alpha\beta)$ for all triples of pair-wise distinct $i, j, k \in \{1, \dots, n\}$. In other words, for $n \geq 3$, every elementary matrix of determinant 1 can be written as a commutator. Since every matrix is expressible as a product of elementary matrices, and commutators are closed under multiplication, we thus obtain $\mathrm{SL}_n(\mathbb{R}) \subseteq [\mathrm{GL}_n(\mathbb{R}), \mathrm{GL}_n(\mathbb{R})]$ for all $n \geq 3$.

The case $n=2$ will be treated in class. ■

9: Consider the group $G := (\mathbb{Z}/8\mathbb{Z}) \times (\mathbb{Z}/4\mathbb{Z}) \times (\mathbb{Z}/4\mathbb{Z})$.

(a) Please find a homomorphism $f : G \rightarrow G$ with $\text{Ker}(f) = \langle (1, 2, 1) \rangle$.

(b) What is the isomorphism type of $G/\text{Ker}(f)$?

Part (a):

Solution #1 (Blitz): First note that

$$\langle (1, 2, 1) \rangle = \{(a, b, c) \in (\mathbb{Z}/8\mathbb{Z}) \times (\mathbb{Z}/4\mathbb{Z}) \times (\mathbb{Z}/4\mathbb{Z}) \mid c = a \text{ mod } 4 \text{ and } b = 2a \text{ mod } 4\}.$$

In other words, we can parametrize the kernel in terms of the first coordinate. Clearly then, the preceding conditions are equivalent to $c - a = 0 \text{ mod } 4$ and $b - 2a = 0 \text{ mod } 4$. So it makes sense to use $f(a, b, c) := (0, b - 2a, c - a)$. ($f(a, b, c) := (0, c - a, b - 2a)$ would also work perfectly well: all that matters is that the two last coordinates be elements of $\mathbb{Z}/4\mathbb{Z}$, thus giving the mod 4 reduction we need.) Indeed, (a, b, c) lies in the kernel of $f \iff (a, b, c)$ lies in the set we just described above.

Solution #2 (A Generalizable Linear Algebra Approach): Let us, for the moment, forget that we are reducing mod 8 or mod 4. It is then easy to obtain all multiples of $(1, 2, 1)$ as the right null-space of a matrix.

More precisely, the 1×3 matrix $\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$ is already in row-echelon form, yielding exactly 2 free variables. So a basis for the right null-space of $\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$ is $\left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \right\}$

However, rearranging our last two basis vectors into a matrix, we also clearly obtain that the left-null space of $\begin{bmatrix} -1 & -2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$ is generated by $\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$. Equivalently, the kernel of the map $f(a, b, c) := (0, c - a, b - 2a)$ is generated by $\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$. (Note that we could have also said the same about the maps $g(a, b, c) := (c - a, b - 2a)$ or $h(a, b, c) := (0, b - 2a, 0, c - a, 0)$: what matters most is that the two linear forms $c - a$ and $b - 2a$ vanish.) The reason we chose f as specified is so that we can then reduce the first, second, and third coordinates respectively by 8, 4, and 4 so that the image lies in $G = (\mathbb{Z}/8\mathbb{Z}) \times (\mathbb{Z}/4\mathbb{Z}) \times (\mathbb{Z}/4\mathbb{Z})$. So we have our desired homomorphism.

Note: The advantage of this solution is that it immediately generalizes to finding homomorphisms from $(\mathbb{Z}/n_1\mathbb{Z}) \times \cdots (\mathbb{Z}/n_k\mathbb{Z})$ to $(\mathbb{Z}/n_1\mathbb{Z}) \times \cdots (\mathbb{Z}/n_\ell\mathbb{Z})$ with specified kernel. We'll do more examples later in class...

Part (b): Since $\langle (1, 2, 1) \rangle$ is a normal subgroup, thanks to its expression as a kernel from Part (a), we have $\#(G/\langle (1, 2, 1) \rangle) = \frac{\#G}{\#\langle (1, 2, 1) \rangle}$. It is then easily checked (e.g., by enumerating all elements) that $\#\langle (1, 2, 1) \rangle = 8$, so we obtain $\#(G/\langle (1, 2, 1) \rangle) = \frac{128}{8} = 16$.

To conclude, we can invoke the Classification Theorem for Finite Abelian Groups (and the fact that $G/\langle (1, 2, 1) \rangle$ is a quotient of $(\mathbb{Z}/8\mathbb{Z}) \times (\mathbb{Z}/4\mathbb{Z}) \times (\mathbb{Z}/4\mathbb{Z})$) to deduce that $G/\langle (1, 2, 1) \rangle$ must be isomorphic to one of the following groups:

$$(\mathbb{Z}/8\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z}), (\mathbb{Z}/4\mathbb{Z}) \times (\mathbb{Z}/4\mathbb{Z}), (\mathbb{Z}/4\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z}).$$

To determine which it is, among many ways, we can look at the order of elements in $G/\langle (1, 2, 1) \rangle$ to try to rule out some of the possibilities.

First note that, $4 \cdot (a, b, c) \equiv (4a, 4b, 4c) \equiv (4a, 0, 0) \equiv (4a, 8a, 4a) \pmod{\langle(1, 2, 1)\rangle}$, for *any* $(a, b, c) \in G$. This means that all elements of our quotient have order dividing 4, and thus there are no elements of order 8. This leaves only two possibilities for our quotient.

Next, observe that $(1, 0, 0)$ and $(2, 1, 0)$ each have order 4 in $G/\langle(1, 2, 1)\rangle$, since $2 \cdot (1, 0, 0) = (2, 0, 0)$ and $2 \cdot (2, 1, 0) = (4, 2, 0)$, neither of which lies in $\langle(1, 2, 1)\rangle$. (So order dividing 2 for these elements is impossible.) Furthermore, it is easily checked that $(1, 0, 0) \not\equiv (2, 1, 0) \pmod{\langle(1, 2, 1)\rangle}$, and $(2, 0, 0) \not\equiv (4, 2, 0) \pmod{\langle(1, 2, 1)\rangle}$. So the subgroups $\langle(1, 0, 0)\rangle$ and $\langle(2, 1, 0)\rangle$ of $G/\langle(1, 2, 1)\rangle$ are disjoint and each has order 4. This is impossible for $(\mathbb{Z}/4\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$, so our quotient is thus isomorphic to $\boxed{(\mathbb{Z}/4\mathbb{Z}) \times (\mathbb{Z}/4\mathbb{Z})}$.